

Trilok Navya Dubaka Data Scientist

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Summary

Results-driven Data Scientist with 4+ years of experience in designing, developing, and deploying machine learning models and analytical solutions. Expertise in Python, SQL, and advanced ML frameworks like TensorFlow, PyTorch, and Scikit-learn. Skilled in cloud platforms (AWS SageMaker, Lambda, Glue, Redshift, GCP), big data tools (Apache Spark, PySpark, Kafka), and ETL pipelines. Adept in data visualization with Tableau, AWS QuickSight, and Power BI. Proficient in sentiment analysis, fraud detection, customer segmentation, and recommendation systems. Experienced in creating scalable pipelines and delivering actionable insights for business optimization.

Technical Skills

- Programming Languages: Python, Java, SQL
- Machine Learning & AI: XGBoost, Random Forest, Isolation Forest, Autoencoders, CNN, RNN, LSTM, SVM, Decision Trees, K-Means Clustering, Transformer Models, Scikit-learn, TensorFlow, PyTorch
- Big Data & Data Engineering: Apache Spark, PySpark, Hadoop, Hive, Apache Kafka, ETL Pipelines, Airflow
- Cloud Platforms: AWS SageMaker, AWS Lambda, AWS Glue, AWS Redshift, AWS Elastic Beanstalk, AWS Kinesis, GCP
- Data Visualization & Reporting: Tableau, AWS QuickSight, Power BI, Matplotlib, Seaborn
- APIs and Automation: Flask, Tweepy
- Databases: MySQL, PostgreSQL, MongoDB, Amazon Redshift, MS SQL Server, NoSQL
- Other Tools & Technologies: Docker, Kubernetes, Jenkins, JIRA, ServiceNow, Pandas, NumPy, PyOD

Professional Experience

Data Scientist, Toshiba

06/2024 – Present | Remote, USA

- Developed and deployed fraud detection models using XGBoost, Random Forest, and Isolation Forest in Python with Scikit-learn and TensorFlow, achieving 95% accuracy in real-time detection and reducing fraud-related financial losses by 30%.
- Utilized AWS SageMaker for model training, hyperparameter tuning, and deployment of machine learning models, while leveraging AWS Lambda for real-time fraud detection, ensuring scalable and low-latency processing of millions of retail transactions.
- Integrated AWS Kinesis to stream high-velocity transaction data for real-time processing, applying anomaly detection techniques and utilizing Python's PyOD library for outlier detection, reducing false positives and enhancing model accuracy.
- Designed efficient ETL pipelines using AWS Glue to transform and prepare transaction data stored in Amazon S3 for machine learning training, improving data preprocessing time by 40% and ensuring consistent, clean data for fraud detection.
- Implemented feature engineering using SQL for complex transaction and customer data extraction, alongside Python libraries like Pandas and NumPy, incorporating customer behavior, time-series analysis, and purchasing patterns into fraud detection models.
- Focused on financial transaction data for high-value retail products, such as electronics and luxury goods, monitoring purchase trends, payment methods, and account behaviors to detect anomalous activity and prevent fraudulent transactions.
- Deployed fraud detection models in production using AWS Elastic Beanstalk, and monitored model performance with AWS CloudWatch, integrating dashboards built with AWS QuickSight for stakeholders to track fraud trends and KPIs in real-time.

Data Scientist, Genpact

07/2020 – 07/2022 | India

- Designed and deployed a sentiment analysis pipeline using Python, NLTK, and Transformers to process and classify over 500,000 customer feedback entries, improving product insights for data-driven decision-making.
- Leveraged Tweepy API to extract real-time customer feedback from social media platforms, integrating data with internal CRM systems using SQL for seamless storage, query execution, and data consistency.
- Developed and fine-tuned sentiment classification models using BERT and Scikit-learn, achieving 90% accuracy; optimized hyperparameters with GridSearchCV to enhance model performance for precise customer sentiment identification.
- Visualized sentiment trends and actionable insights through interactive dashboards in Tableau and Power BI, enabling cross-functional teams to track customer feedback metrics and prioritize product feature enhancements effectively.
- Automated data ingestion, preprocessing, and model deployment workflows with Apache Airflow and AWS Lambda, ensuring a scalable and reliable sentiment analysis system monitored via AWS CloudWatch.

Junior Data Scientist, HCL Tech

07/2018 – 07/2020 | India

- Assisted in building a recommendation engine using collaborative filtering and content-based models with Python (scikit-learn) to analyze customer purchase patterns, increasing engagement and product visibility by 15%.
- Performed data preprocessing and feature engineering on user activity data (clickstream and purchase history) using Python (Pandas, NumPy) and SQL, ensuring clean and accurate inputs for model training and evaluation.
- Supported large-scale data processing by implementing ETL workflows on AWS Glue and optimizing data pipelines with PySpark, improving data readiness for real-time recommendations by 25%.
- Deployed machine learning models using Flask APIs for integration with e-commerce platforms, enhancing the system's real-time response to customer interactions under 300ms.
- Created interactive dashboards in Power BI, visualizing recommendation performance metrics like click-through rates and customer retention, enabling actionable insights for cross-functional teams and improving strategic decision-making.

Education

Master of Science, Arizona State University

08/2022 - 05/2024 | Arizona, USA

Information Technology

Graduate Teaching Assistant - Unix/Linux and Python

Bachelor of Technology, JNTU

06/2013 - 05/2017 | Hyderabad, India

Computer Science and Engineering

Projects

Recommendation Engine for Ad Campaigns (Python, Pandas, Numpy, Regression, Scikit-learn) [\[GitHub\]](#)

- Built a recommendation system to suggest products based on purchase patterns, achieving 0.98 accuracy using logistic regression.

Retail Business Customer Segmentation (Pandas, Clustering, Feature Engineering, Matplotlib, K-Means) [\[GitHub\]](#)

- Clustered retail customers using K-Means and RMF analysis to derive behavioral insights.

Customer Churn Prediction (Pandas, Decision Tree, Business Metrics) [\[Github\]](#)

- Predicted customer churn using Decision Tree Algorithm and analyzed business metrics like churn rate and retention rate.